

25 June 2024

Ref: CL/4475

Subject: **First consultation on the implementation of the 2021 Recommendation on Open Science**

Madam/Sir,

Under Article VIII of UNESCO's Constitution, Member States are required to submit regular reports on the measures they have taken to implement the conventions and recommendations adopted by the Organization.

In this context, I invite your Government to submit its report on the 2021 Recommendation on Open Science, in English or French, **no later than 28 February 2025**. Attached you will find guidelines to assist your authorities in submitting the report. Regional online sessions will also be organized to provide further guidance on report preparation.

Member States are encouraged to organize all necessary consultations with relevant ministries and institutions as well as with key stakeholders, including professional associations, civil society partners, private sector entities and National Commissions for UNESCO.

In addition, Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO regarding reports on the 2021 Recommendation on Open Science. Member States should inform the Secretariat of this designation by 20 July 2024 at the latest at: openscience@unesco.org.

The Secretariat will present the first global report on Member States' application of this Recommendation for examination at the 222nd session of the Executive Board in autumn 2025. This report, together with the comments of the Committee on Conventions and Recommendations, will subsequently be submitted to the 43rd session of the General Conference in 2025.

The Natural Sciences Sector remains at your disposal for any further information you may require.

Please accept, Madam/Sir, the assurances of my highest consideration.

Audrey Azoulay
Director-General

Encl.: *Guidelines for the preparation of the reports from Member States on the implementation of the 2021 Recommendation on Open Science*

cc: Permanent Delegations to UNESCO
National Commissions for UNESCO
UNESCO regional offices
UNESCO field offices

To Ministers responsible for relations with UNESCO

ANNEX

GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i.e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on the extent to which Member States have integrated the core values and guiding principles of open

science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted [online](#).
- (ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: **TANZANIA**
- Organization(s) or entity(ies) responsible for the preparation of the report*:

Name: **Ministry of Education Science and Technology**

Website: **www.moe.go.tz**

Email address*:

Phone number*:

Please describe the role/mandate of your organization: The ministry is obligated to coordinate research and development in science and technology, develop of local expertise in science technology and innovation and promote application of science, technology, engineering and mathematics across the country.

- Officially designated contact person(s) who completed this survey:

Full name*: **Bahati E. Kimario**

Position*: Officer under the directorate of Science, Technology and Innovation

Email address*:

- Full Name(s) of designated official(s) certifying the report: **Prof. Ladslaus Mnyone**
- Brief description of the consultation process established for the preparation of the report: **Due to time constraints, a similar questionnaire was developed and distributed via Google Forms to 68 institutions, which includes Research and Development institutions, Regulatory Authorities, Universities, and Startup Associations from both public and private sectors. Of these, 20 institutions responded, and the analysis of their responses formed the basis for the development of this report.**

Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

S/ N	Name of the organization	Website	Email address	Sector
1.	Nelson Mandela African Institution of Science and Technology	https://nm-aist.ac.tz/		Public
2.	Tanzania Atomic Energy Commission	https://www.taec.go.tz/		Public
3.	Commission Science and Technology	https://costech.or.tz/		Public
4.	Mbeya University of Science and Technology	https://must.ac.tz/		Public
5.	Tanzania Bureau of Standards (TBS)	https://www.tbs.go.tz/		Public
6.	University of Dar es Salaam	https://www.udsm.ac.tz/		Public
7.	Tanzania Fisheries Research Institute (TAFIRI)	https://www.tafiri.go.tz/		Public
8.	University of Iringa	https://uoi.ac.tz/		Private
9.	Tanzania Forestry Research Institute (TAFORI)	https://tafori.or.tz/		Public
10.	Tanzania Gemological Centre	https://tgc.ac.tz/		Public

11.	Kampala International University in Tanzania	https://kiut.ac.tz/		Private
12.	Dar es Salaam University College of Education	https://duce.ac.tz/		Public
13.	Tanzania Automotive Technology Centre	https://www.tatcnymbu.or.tz/		Public
14.	Tanzania Industrial Research and Development Organisation	https://www.tirido.or.tz/		Public
15.	Tanzania Forestry Research Institution	https://tafori.or.tz/		Public
16.	Tanzania Meteorological Authority	https://www.meteo.go.tz/		Public
17.	Kairuki University	https://www.hkmu.ac.tz/		Private
18.	Dar es Salaam Maritime Institute (DMI)	https://www.dmi.ac.tz/		Public
19.	Mwalimu Nyerere University of Agriculture and Technology (MNUAT)	https://www.mjnuat.ac.tz/		Public
20.	Mining Commission	https://www.tumemadini.go.tz/		Public

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES / NO

Yes, the Ministry has disseminated and promoted the 2021 recommendations to relevant ministries and institutions through various workshops and seminars. For instance, in March 2023, the Ministry of Education, Science and Technology (MoEST), through the Tanzania Commission for Science and Technology (COSTECH), organized

a National Open Science Dialogue to explore the challenges and opportunities associated with establishing an open science ecosystem in Tanzania. This dialogue underscored the need for a national open science policy to guide research dissemination, data sharing, and incentivize open science practices. Furthermore, in December 2024, MoEST conducted an open science awareness-raising seminar for staff members of universities and Technical and Vocational Education and Training (TVET) institutions.

- 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

- 1.3 Have there been awareness raising activities on open science, including all its key elementsⁱ, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

YES / NO

If yes, please provide details on the activities organized or foreseen and links as relevant.

Yes, there have been awareness-raising activities on open science, although the landscape of such initiatives varies across institutions. The Ministry of Education, Science and Technology (MoEST) has organized awareness-raising workshops for Higher Learning Institutions (HLIs). Additionally, institutions such as the Nelson Mandela African Institution of Science and Technology (NM-AIST), the Mining Commission, the Tanzania Fisheries Research Institute (TAFIRI), the Tanzania Gemological Centre (TGC), and the Tanzania Meteorological Authority (TMA) have actively conducted internal workshops and training sessions to promote open science practices.

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

YES / NO

Several institutions have undertaken concrete measures to integrate open science principles into publicly funded research. For example, the Tanzania Commission for Science and Technology (COSTECH) has established open-access institutional repositories and requires researchers to publish their findings in open-access journals, thereby promoting transparency and accessibility. Similarly, the Tanzania Gemological Centre (TGC) has incorporated open science practices into its research funding mechanisms, supported by government funding for infrastructure development. In addition, the Nelson Mandela African Institution of Science and Technology (NM-AIST) encourages researchers to adhere to open science principles when applying for funding. A notable example is the recently submitted proposal titled “*FAIR-DATASHARE: Enabling Findable, Accessible, Interoperable, and Reusable (FAIR) Research Data Sharing among Higher Education Institutions (HEIs) Researchers in Tanzania Using Research Data Repositories (RDRs)*,” submitted to the VLIR-UOS Short Initiative Call 2025. Furthermore, institutions such as Kampala International University in Tanzania (KIUT) have adopted open science practices as a prerequisite for funding, particularly for health-related research funded by the European Union.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?

YES / NO

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

YES / PARTLY / NO

- 2.3 In your country, are there specific policy instruments^v that aim to promote open science?

YES/ UNDER DEVELOPMENT/NO

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\) 17.a, 17.b, 17.c, 17.f, 17.h](#))

YES / UNDER DEVELOPMENT / NO

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(iii\) 17.d, 17.e, 17.g](#))

YES / UNDER DEVELOPMENT/ NO

Some institutions, like Tanzania Commission for Science and Technology (COSTECH) and Kampala International University in Tanzania (KIUT), have established policies and strategies aligned with the 2021 UNESCO Recommendation on Open Science. COSTECH, for instance, has developed a National Data Sharing Framework for Science, Technology and Innovation and promotes participatory science through public involvement in research projects. Similarly, KIUT's research policy emphasizes community involvement and participatory research. However, many institutions operate under broader national policies that but lack standalone institutional policies. For example, TGC aligns with Tanzania's National Research and Development Policy but does not have a specific open science policy.

- 2.6 In your country, are there specific funding mechanisms^{vi} for open science at the national level? (ref.: [\(ii\) 17.j, \(iii\) 18.j, \(iv\) 19.c](#))

YES / UNDER DEVELOPMENT / NO

- 2.7 In your country, have open science practices^{vii} been integrated in existing research funding mechanisms?

YES / NO

Some, institutions like COSTECH and KIUT have successfully integrated open science practices into their research funding mechanisms, emphasizing open-access publishing and data sharing. COSTECH's funding guidelines require researchers to disseminate their findings through open-access platforms, promoting transparency and collaboration. Similarly, KIUT's research funding, particularly from the European Union, mandates open science practices as a precondition for funding.

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

YES / NO

Efforts to foster public-private partnerships (PPPs) on open science are underway. Institutions like TGC and COSTECH, which have signed MoUs and engaged in collaborative projects with private sector organizations. TGC, for example, has partnered with the Gemmological Association of Great Britain (Gem-A) to enhance gemological education and research. Similarly, COSTECH's strategic plan emphasizes increasing public-private partnership engagement in science, technology, and innovation.

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.j, 23.c](#))

YES / UNDER DEVELOPMENT / NO

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

No official data

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

According to the Tanzania Communications Regulatory Authority (TCRA) Communication Statistics Report of December 2024, 91% of the population was covered by 3G services, and 88% by 4G services. In terms of geographical coverage, 3G and 4G services reached 74% and 71% of the country's area, respectively. Additionally, 5G services became available to 20% of the population.

- 3.3 Are there national research and education networks (NRENs)^{viii} active in the country? (ref.: [\(iii\) 18.c](#))

YES / NON

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

<https://costech.or.tz/>

- 3.4 Are there national or institutional [open science infrastructures](#)^{ix} in your country? (ref.: [\(iii\) 18.d, 18.e, 18.k, 18.l, 18.m](#))

YES / UNDER DEVELOPMENT / NO

Some institutions, like COSTECH and TAFIRI, have developed open science infrastructures, such as high-performance computing (HPC) and electronic data systems, to support open science practices. COSTECH's HERIN provides high-speed internet connectivity and facilitates access to digital libraries and research databases.

Similarly, TAFIRI has developed the Electronic Catch Assessment System (eCAS) for data collection and sharing.

- 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: [\(iii\) 18.e](#), [18.f](#))

YES / NO

Some institutions such as COSTECH, hosts federated IT infrastructures (HERIN) which promote open science by providing high-speed internet connectivity and access to digital libraries. Similarly, Tanzania Atomic Energy Commission (TAEC) hosts the International Nuclear Information System (INES) under the International Atomic Energy Agency (INES)

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: [\(iii\) 18.g](#))

YES / NO

Some institutions like COSTECH and TMA are actively engaged in North-South and South-South collaborations to optimize infrastructure use and promote open science. COSTECH, for example, collaborates with the National Research Foundation (NRF) of South Africa and participates in the O.R. Tambo Africa Research Chairs Initiative. Similarly, TMA collaborates with the World Meteorological Organization (WMO) and regional climate initiatives like CORDEX Africa.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a](#), [19.c](#))

YES / NO

Some institutions, like COSTECH and KIUT, have implemented systematic capacity-building initiatives, including workshops and training sessions on open science. COSTECH, for example, has launched national frameworks and research chairs programs to enhance research and innovation. Similarly, KIUT provides training programs and infrastructure support for open science practices.

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES / NO

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

YES / NO

Few institutions, like KIUT, have incorporated open science competencies into their curricula. Also, the University of Dar es Salaam (UDSM) is planning to incorporate open science competencies into its curricula through the ERASMUS CBHE project

- 4.4 Are open educational resources^x as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

YES/ **PARTLY**/ NO

Institutions reported minimal usage of Open Educational Resources (OER) to advance Open Science. MNUAT and the University of Iringa are beginning to acknowledge the potential of OER. TGC and TBS have not yet developed or implemented OER initiatives in line with the 2019 UNESCO Recommendation

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))

YES / NO

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b, 20.c, 20i](#))

YES / NO

KIUT, have reviewed their research assessment and evaluation processes to align with open science principles.

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.c, 20.d](#))

YES / NO

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h](#))

YES / NO

Some institutions, like KIUT and TAFIRI, have implemented initiatives to recognize and reward open science practices.

- 5.4 Have there been any unintended negative consequences^{xi} of open science practices reported in your country? (ref.: [\(v\) 20](#))

YES / NO

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21](#))

YES

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

- 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a](#))

Tanzania is actively promoting and facilitating international scientific collaborations on open science, aligning with the 2021 UNESCO Recommendation on Open Science. The Tanzania Commission for Science and Technology (COSTECH) is leading efforts to develop a national open science policy that harmonizes existing policies and provides a cohesive framework for open science practices across the country. Collaborations with organizations like the Consortium of Tanzania Universities and Research Libraries (COTUL) and Electronic Information for Libraries (EIFL) have been instrumental in these developments. Additionally, COSTECH has organized training workshops to enhance researchers' skills in securing research funding and adopting open science practices, including participants from both Mainland Tanzania and Zanzibar. Efforts are also underway to improve open science infrastructure, such as establishing institutional repositories and adopting open journal systems, to make research outputs more accessible and promote data sharing among researchers. Furthermore, Tanzania has participated in regional dialogues, such as the East African Community's initiatives on open science, to address challenges and define strategies for implementing open science practices. Through these concerted efforts, Tanzania is making significant strides in aligning with the UNESCO Recommendation on Open Science, fostering an environment conducive to international scientific collaboration and the advancement of open science principles.

- 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b, 22.e, 22.f](#))

YES / NO.

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/page=21>)

YES / NO

Tanzania is actively engaged in discussions and initiatives aimed at establishing regional and international funding mechanisms to support open science, in alignment with the 2021 UNESCO Recommendation on Open Science. As highlighted above, in February 2023, a High-Level Multi-Sectoral National Open Science Dialogue was convened in Tanzania, bringing together stakeholders from academia, research institutions, and policy-making bodies. This dialogue, co-hosted by the Tanzania Commission for Science and Technology (COSTECH), the East African Science and Technology Commission (EASTECO), the Public Library of Science (PLOS), and the Training Centre in Communication Africa, focused on strategies to promote open science and explored avenues for securing funding from both governmental and international partners. Additionally, Tanzania's participation in regional collaborations, such as the African Union (AU), East African Community (EAC), and Southern African Development Community (SADC), underscores its commitment to fostering international cooperation and mobilizing resources to advance open science initiatives.

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

8. GENERAL CONSIDERATIONS

- 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

Financial constraints often limit funding for open science, as many funders continue to prioritize traditional research models. Inadequate digital infrastructure, particularly unreliable internet access, further hinders the effective implementation of open science practices. Additionally, existing national and regional policies may not fully support open science, creating regulatory obstacles. Some researchers are also hesitant to adopt open approaches due to concerns over intellectual property rights and data privacy. Compounding these challenges is a general lack of awareness and training on the benefits of open science, highlighting the need for targeted capacity building and support for researchers

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- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
- ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
- iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
- iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
- v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
- vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
- vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
- viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
- ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

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- x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
- xi For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.